

Project Brainstorming

Project Brainstorming: [Project Name]

Code to Give

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Context

Passion to Serve is a volunteer-run nonprofit in Singapore that supports migrant workers through distribution of essential supplies and enrichment programs such as wellness and digital literacy. Each event requires complex coordination across volunteers, venues, logistics partners, and beneficiaries currently managed through multiple disconnected systems. As a result, time and energy that could be spent supporting migrant workers is diverted to managing logistics, limiting Passion to Serve's ability to deliver and scale their impact.

Problem Statement

How might Passion to Serve create a simple, unified way to coordinate volunteer-led events so that organizers can easily track progress, manage volunteers, and execute events with confidence?

Mission

Build a centralized event coordination tool that helps Passion to Serve plan, manage, and execute volunteer-led events from start to finish. The solution should:

- Centralize all event coordination into a single view for each event
- Provide clear visibility into responsibilities, progress, and outstanding tasks
- Enable organized communication across volunteers, organizers, and external partners

- Simplify volunteer management including registration, communication, attendance tracking, and post event follow ups
- Make key event information easy to find, update, and share
- Reduce reliance on fragmented tools and manual coordination

Target Audience

- **Migrant Workers:** Primary beneficiaries seeking essential supplies, enrichment, and wellness support.
- **Volunteers:** Individuals contributing time as educators or event coordinators.
- **Organization Admins:** Staff managing logistics, partners, and data entry.

User Stories

Organiser

- As an Organiser, I would want to know how many participants and volunteers are coming, so that i can plan for manpower constraints with the logistics team
 - I would also want logistic teams to be able to view these numbers
- As an Organiser, I want to see all tasks for an event on one board with beneficiaries and deadlines so I know what is outstanding without checking the different chats
- As an organiser, I want a simple workflow to send out announcements to volunteers without having to email/text each volunteer 1 by 1.
- As an organiser, I want the system to automatically message backup volunteers if headcount falls short, so that I don't have to manually chase people
- As an Organiser, I want a tool to suggest specific next action when readiness score is low (e.g. re-invite past volunteers) so I know what to do
- As an organiser, I want to have templates on past repeating events, so that I am able to refer/repeat the same actions taken to organise the event.
- ** (calendar/whentomeet UX) As an organiser, I want the selection of a feasible date and time to organise transport logistics/ recreation/distribution drive (25-30 volunteers per event)
- ** As an organiser I want to move away from excel sheets and into a more centralised database especially since I have more manpower and events to manage

- As an organiser, I want to be able to have a template that shows which logistic partners are required for specific events, so that i have a consolidated view of who to contact

Volunteer

- As a volunteer, I would want to be notified of events, so that I can participate in those I am interested in
- As a volunteer, I want to be able to join a waitlist which adds me into the event if somebody pulls out and notifies me when a spot opens so I am automatically added, and given notice
- As a Volunteer, I would want to be able to view all available events in one page
- As a volunteer, I would want a to be able to sign up for events in the press of a single button, so that I skip filling out a separate form each time
- As a Volunteer, I would want to view the events in calendar format, so that i can plan my schedule
- As a Volunteer, I would want to be notified before an event, so that I do not miss the event
- ** As a Volunteer, I want to confirm my participation just by replying “yes” on WhatsApp, so that I don’t have to fill out a separate form every time
- As a volunteer, I want to receive suggestions to an event I’m likely to enjoy; so that I don’t have to scroll through generic broadcast posts (ML)
- ~~● As a volunteer, I want to have a positive experience so I would want to continue volunteering~~
- ~~● As a new volunteer, I would like additional guidance on how to approach things for the first time and learn about how things are done~~

Beneficiary

- ** As a Beneficiary, I would want to be notified of events, so that I can participate in those I am interested in.
- As a Beneficiary, I want to be able to join a waitlist which adds me into the event if somebody pulls out and notifies me when a spot opens so I am automatically added, and given notice
- ** As a beneficiary, I would like a support group I could reach out to
- As a beneficiary, I would like to receive reminders of where I left off in my educational modules and to continue on (Past history on classes attended)
- As a Beneficiary, I would want to be able to view events with clear details easily on an app, so that I know what the event is about, what to bring etc.

Key Features / Requirements

Category	Feature Name	Description	Name	Status	Remarks
Centralized Volunteer & Beneficiary Management	Centralized Database (Single Source of Truth)	Replaces the current spreadsheets and disconnected chats with one database storing every volunteer's and beneficiary's contact info, role/skill tags, event history, and hours — queryable by any organizer instead of living in separate files.	Zhi Song		Missing Tables, Database for both volunteers and beneficiaries required.
	New Volunteer Onboarding Flow	First-time volunteers automatically receive a short WhatsApp/app sequence covering what to expect, event etiquette, who to contact on-site, and a pre-event checklist — before their first shift, without an organizer manually briefing them.	Zhi Song		
	Smart Waitlist & Auto-Backfill	When an event fills up, further sign-ups go to a waitlist. If a confirmed volunteer or beneficiary cancels, the system automatically promotes the next person on the waitlist and sends them an instant WhatsApp notification with a Confirm/Decline reply option.	Nickson		
Event Planning & Logistics Coordination Dashboard	Live Headcount Dashboard	A real-time counter per event showing confirmed volunteers, waitlisted volunteers, and confirmed beneficiaries. A read-only version of this link can be shared directly with transport/logistics partners so they can plan without needing platform access.	Chermaine	Waiting on Zhi Song's Database	
	Unified Event Task Board	A Kanban board (To Do / In Progress / Done) per event where organizers assign tasks with owners and deadlines — so "what's outstanding" is visible in one place instead of scattered across WhatsApp threads.	Chermaine		
	Event Templates & Duplication	Any past event (e.g. "GIFTIK Distribution Drive") can be saved as a template capturing its task list, logistics-partner checklist, and volunteer roles. Organizers clone it in one click for the next occurrence and just update the date/venue.	Aditi		
	Hoto Dashboard	A dashboard for incoming volunteers to view and check on the progress of the current logistics..	Aditi		

Category	Feature Name	Description	Name	Status	Remarks
Communication & Notification System (WhatsApp / Green API)	One-Click Volunteer Broadcast	Organizers write one announcement and send it to all volunteers, or a filtered subset (e.g. only those confirmed for Event X), via WhatsApp — instead of messaging people individually.	Zhi Song	✓	
	WhatsApp RSVP Bot	Volunteers and beneficiaries confirm attendance by replying "YES"/"NO" to an automated WhatsApp message; the bot parses the reply and updates their status in the database instantly — no separate form to fill out.	Zhi Song	In Progress	
	Automated Event Reminders	Scheduled WhatsApp reminders fire automatically at set intervals before an event (e.g. 3 days out, 1 day out) with venue, start time, and what to bring. Tied with “event detail page”	Zhi Song		
	Automated Backup-Volunteer Alerts	If a confirmed headcount for an event drops below its target a set number of days out, the system automatically messages past volunteers tagged as "available/backup" — so the organizer isn't manually chasing people down.	Nickson		
	Change Date/Time on Promotional Video	KIV	KIV		
Event Discovery & Registration	Event Explorer Page	One scrollable, filterable page listing every upcoming event with type, date, location, and spots remaining, for volunteers and beneficiaries to browse.	Aditi		
	Calendar View	Toggle to see all registered (or all open) events laid out on a monthly/weekly calendar, so volunteers can plan their schedule at a glance.	KIV		
	Detailed Event Info Cards	Each event has a detail page covering what the event is, what to bring, dress code, meeting point, and duration — so beneficiaries know exactly what to expect before showing up.	Zhi Song		- json(output) -
Digital Certification & Learning Progress Tracking	Course Progress Tracker	For beneficiaries in multi-session courses (e.g. Spoken English), the system shows which sessions they've completed and what's next, with a nudge if a session is missed.	KIV		

Category	Feature Name	Description	Name	Status	Remarks
	Digital Certificates	On completing a course/assessment, beneficiaries automatically receive a shareable digital certificate; volunteer educators similarly receive participation certificates for sessions they ran.	KIV		
Smart Insights & Automation	Event Readiness Score	A single score (0–100%) per upcoming event, computed from confirmed headcount vs. target, task-board completion %, and logistics-partner confirmation status — giving organizers an at-a-glance health check.	Chermaine		
	Suggested Next Actions	When the readiness score is low, the system surfaces a concrete action instead of just a red number — e.g. "Re-invite these 12 past volunteers" or "3 logistics tasks are overdue — nudge [assignee]."	Chermaine		
	Personalized Event Recommendations	A lightweight model, based on each person's past event types attended, surfaces 2–3 suggested upcoming events per volunteer/beneficiary instead of one generic broadcast list — increasing relevant sign-ups.			
Retention		Volunteer Loyalty programme - where the technological solution can track involvement of a Volunteer with Passion To Serve over time and give credits/loyalty points.	Girija		
		sending same/repeat welcome messages and event information to repeat volunteers as Excel does not know that they had volunteered with us in the past as well.	Nickson		Create DB for volunteers and beneficiaries to determine if they have already volunteered
		track each beneficiary enrolled with us for their history of courses/events attended with us. VIP PASS (kind of loyalty programme) for the migrant workers who had been Passion To Serve's past students and they were given VIP access to our sought-after GIFITK event (which can see queues going into hundreds) - it was challenging to process all this through Excel.	Girija		

Required Functionality

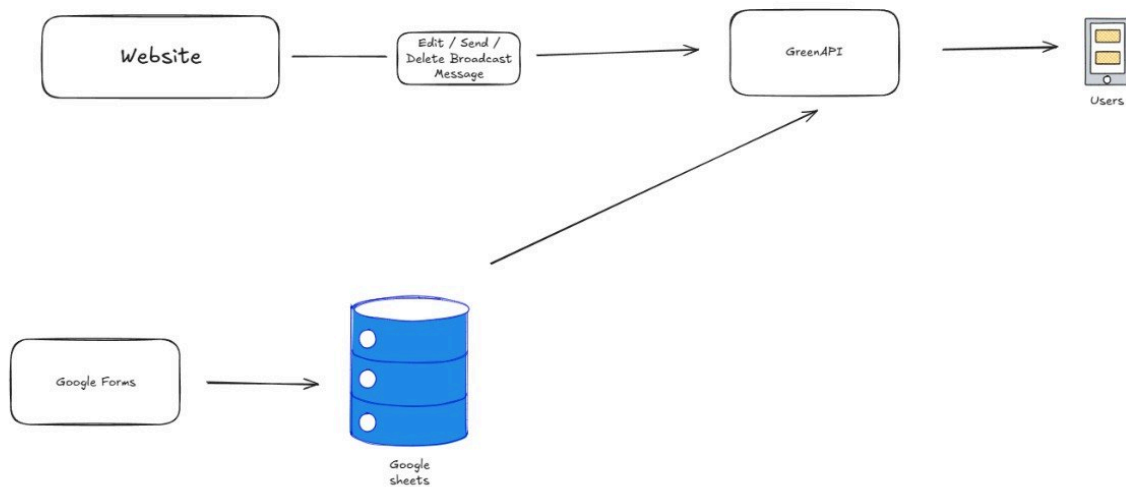
- Centralized Volunteer Management System
- Automated Messaging and Notification System for Beneficiaries
- Digital Certification and Progress Tracking for Courses
- Venue and Logistics Coordination Dashboard

Technical Architecture

[Define the technology stack and system design]

- **Frontend:** [Framework/Library choice]
- **Backend:** [API/Server framework choice]
- **Database:** [Storage solution]
- **Authentication:** [Auth provider/method]
- **Integrations:** [APIs or third-party services required]

Green API architecture:



Development Roadmap

Phase	Focus Area	Key Tasks
[Phase Name]	[Focus]	[Key deliverables]

Future Implementations

- As an organiser, I want donated items to be automatically flagged if defective, so that I don't have to manually inspect every item myself (CV)
-

Success Metrics

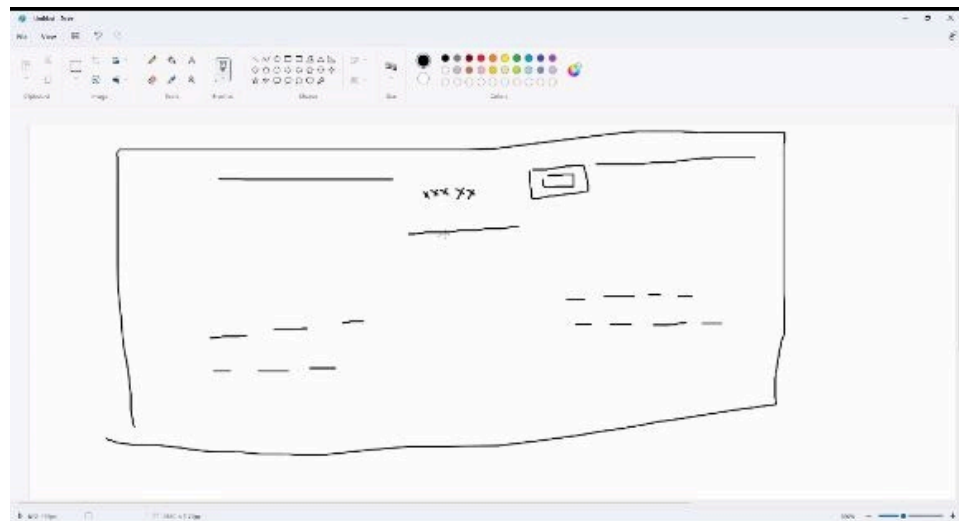
[Define how the project will be evaluated]

- [Metric 1]
- [Metric 2]
- [Metric 3]

Questions

Mentor Feedback

- Play recorded demo.. One team created a mvp and hosted in cloud , on top of recorded..
Have qr code.. Scan and access the website
- Demo 3 mins..
- Team 4 - present at 3:23 (6 minutes)
 1. focused on business use case—profits & feasibility
 2. technical side— why u used this tech stack and not others)
- Judging criteria: equally split(judges .. senior management (Mok guy) / in 30's/40's) + total of 10 teams.. Similar ideas might get boring
- Volunteer- no full-time staff, some volunteers join randomly.. Handover process can be confusing.. No extended SOP, cuz no full time member enforcing.. Not enough resources.. Create a tool to address the concern of full time staff.. Dashboard.. Volunteer A(attends 3, cmi 4th) .. volunteer B(takes over 4th),, instead of a manual handover and get all the information instantly without needing to seek help
- Skit .. (brandon will confirm if he can join).. For demo part
 - Considerations
 - How are judges gg see content? Include content in laptop screen
 - Do a backup recording of demo as a plan B incase anything does work like weak wifi
 - Scenes:
 - 2 rooms will be combined into one big room



- Justify and backup decisions (like why whatsapp or why another app)
- QnA :
 - technical: why u choose this tech stack
 - non-technical: why whatsapp instead of teams ect
- Recommended Website:
 - Has all the good Ui designs compiled from popular websites(he'll share later)--ui first impression.. Judges might go into github

Questions

1. We noticed WhatsApp seems to be your main communication channel. Could you help us understand how central it is to your workflow --> is it your primary tool because most migrant workers are already comfortable with it? And would the team be open to adopting a different platform, or is WhatsApp something you'd want to keep given how the workers use it?
2. That being said, does our solution have to consider specifically migrant workers as a stakeholder?
3. For the solution itself, would a mobile app or a web-based platform work better for how your team and volunteers operate day to day? We'd like to build for whatever fits most naturally into your existing habits.
4. We understand Excel is part of your current process. What is the current issue with it? Is it mainly the manual data entry being time-consuming, or are there limitations with Excel itself? Are there any features you often wish you had that it doesn't offer?
5. Setting aside the venue licensing and booking side of things, is there any other part of the donation drive process that tends to be a pain point or eats up more of your time than it should?
6. Would you rather we build a self-contained solution where everything — chat, information, coordination — lives in one place? Or would you prefer something that integrates with the tools you already rely on, like WhatsApp and Excel, so the team doesn't have to change too much?
7. Could you walk us through how a donation drive actually unfolds on the day? For example, do the workers browse and pick items freely from what's laid out, and roughly how long does the whole event take from start to finish for everyone to collect what they need? Understanding the flow on the ground would really help us design around it. Or do we not need to worry about the actual process of the event itself but rather the set up and the tear down logistically.
8. For the educational courses, we'd love to get a sense of the numbers, roughly how many sign up versus how many complete a course? And for those who don't finish, do you have a sense of what tends to get in the way? Understanding the drop-off would help us think about how to support completion.
9. When it comes to coordinating with your logistics partners, what's the biggest challenge at the moment? For instance, is it difficulty reaching them, consolidating information coming in from several partners at once, or aligning on

a time to meet? We'd like to understand where the coordination tends to break down.

Answers

1. Whatsapp is our main communication tool at the moment. Because:

[A] It is convenient to use (sending posters, images, videos sometimes)

[B] It is used by almost 95% of the population of beneficiaries in Singapore - so it is the most commonly used communication tool. If a beneficiary is not on Whatsapp then we would send them a normal text message and/or an email. So ideally we would want to continue to use Whatsapp unless we can migrate the beneficiaries to our own App which can keep them registered and can send them notifications about course information/event/activities.

Challenge with communicating ONLY through our own App could be about whether beneficiaries would want to install and keep one more App on their mobile phone - may be yes, if they are enrolled with us and are interested in our activities and want regular updates (like they may choose to stay on on our Whatsapp Groups even after a course has been completed)

Challenge with Whatsapp is the Meta's communication and broadcasting policies - which effectively restrict us from broadcasting course information to the registered participants (strangers) in bulk - a solution could be enrol for Meta's or a third party's paid subscription (Whatsapp Business API). Happy to explore all these solutions.

2. We would want to **keep more than just migrant workers as beneficiaries in mind** (could be inmates, could be low income family households, could be members of broader Singapore community - for whom also we can conduct say Free AI or Spoken English course)

3. We would have imagined that a **Mobile App would empower us** to stay closer to our beneficiaries (sitting on their mobile device) including the ability to send notifications about events/activities. Currently every online interaction with volunteers and beneficiaries is through Whatsapp. But yes technically if you feel having an App would be too complicated and challenging from a long term maintenance perspective then we would be open to having a web-based solution as well. A mobile app also gives ease of use and can keep the users logged in - while a web-solution requires them remember one more password.

4. Excel is used as a database and while manual data entry becomes onerous especially when handling large numbers of volunteers and beneficiaries, it does not provide efficient and robust workflow. For instance, it would be good to build a Volunteer Loyalty programme - where the technological solution can track involvement of a Volunteer with Passion To Serve over time and give credits/loyalty points leading to their eventual felicitation/appreciation or even being invited to join Passion To Serve's Executive team - Excel cannot do all this. We also end up sending same/repeat welcome messages and event information to repeat volunteers as Excel does not know that they had volunteered with us in the past as well. Similarly for the beneficiaries, we would want to track each beneficiary enrolled with us for their history of courses/events attended with us. Recently we also started a VIP PASS (kind of loyalty programme) for the migrant workers who had been Passion To Serve's past students and they were given VIP access to our sought-after GIFTIK event (which can see queues going into hundreds) - it was challenging to process all this through Excel. Also with a proper technological solution, we can get new beneficiaries to register with us when they come to us for any recreational or GIFTIK event just by scanning a QR code. The App will remember each beneficiary and immediately know their interaction history with Passion To Serve. We would also want everyone attending our events or courses to become part of our WhatsApp groups and become our Facebook page follower.

5. GIFTIK Donation Drive involves sending pre-set posters to various partner condominiums/HDBs about the next Collection drive. As a separate subset of a solution, we can consider automating PDF/JPG poster generation (just change date/time) and send them to preset emails of the condominium management. Transportation is a logistical hurdle but we would not consider a technological solution as booking a lorry on LalaMove already uses their web/app. **Volunteer management** (enrollment, communicating with them about event information, and then sending them Volunteer certificates after the event) , on the other hand, requires reasonable time investment.

6. We would prefer a self-contained solution however, while we would want to keep integration with Whatsapp (as discussed in #1 above), we can do away with Excel completely if the solution itself provides robust database handling features.

7. Yes, the Workers pick items freely on the day of the Distribution Drive (GIFTIK). The items are sorted and laid out by the Passion To Serve volunteers on the day of the event. The sorting takes about 1.5 hours while the actual event of letting workers flow in and pick items takes about 2 hours and then cleaning and winding up takes about 30 mins.

So workflow around the event: Coordination with Management of Condominium for Collection Drive at Condos, letting residents donate items at designated place at Condominium, Transportation of items from Condo to Recreation Centre, Items unloaded at Recreation Centre and sorted and laid out by Volunteers, Migrant workers queue up, GIFTIK Event opened up for workers to walk in (batch of 20-30 workers at a time) and select 1-3 items at a time and walk out. Would be happy to provide more details and hear any ideas for us to **make this process more efficient**.

8. The numbers can vary depending on the popularity of a topic. For instance, recently our Computer Technology course saw about 375 online registrations (Google form) and about 120 participants came in the first session (that itself was huge), and then about 90 participants stayed on for the rest of the 4-session course. Other smaller courses like financial literacy would see about 100 registrations, but 30 will actually turn up and about 25-30 will eventually complete the course. Usually, once the participants come for the course, they are very likely to complete the course. But the biggest drop happens from the number of online pre-registrations to they actually physically turning up to attend a course - so only say 30-40% of course registrations actually attend the course.

9. Logistics and other partners are:

0. Ministry of Manpower, ACE Group

a. Management of Recreation Centres/Dormitories

b. Management of Condominiums/Apartments

c. Transportation Company such as LalaMove

d. Other NGOs (RSVP, Willing Hearts)

e. Other Education Institutions - for sourcing volunteers

Communication with all of them happens over email - the biggest challenge is really shortage of proper (full/part-time) staff to promptly reply and coordinate with these partners - many times, we face time delays in interacting with these stakeholders. Given the very diverse nature of interactions with all these partners, we are not sure whether and how we can automate or make the process more efficient using a technological solution.

summarised ver

Here's a concise summary of the key requirements and challenges:

Summary of Requirements and Challenges

1. Communication

- **WhatsApp** is the primary communication channel because:
 - It supports images, posters, videos, and documents.
 - Around **95% of beneficiaries in Singapore use WhatsApp**.
 - SMS and email are used only when beneficiaries do not have WhatsApp.
- The long-term vision is to migrate users to a **dedicated platform (mobile app or web app)** while maintaining WhatsApp integration.
- Key challenge:
 - Meta's WhatsApp broadcast restrictions limit mass messaging to participants.
 - Potential solution: **WhatsApp Business API** or a third-party messaging platform.

2. Beneficiary Management

The system should support multiple beneficiary groups, not only migrant workers, including:

- Migrant workers
- Inmates
- Low-income families
- General Singapore community
- Future beneficiary groups

3. Mobile App vs Web Platform

Preferred solution:

- A **mobile app** to:
 - Keep users logged in
 - Send push notifications
 - Increase engagement
 - Act as a permanent communication channel

Alternative:

- A **web-based platform** if mobile app maintenance becomes too costly or complex.

4. Replace Excel with a Central Database

Current reliance on Excel creates significant manual work.

A new system should:

- Maintain volunteer profiles and history
- Maintain beneficiary profiles and participation history
- Track courses and events attended
- Support volunteer loyalty points and recognition programmes
- Support beneficiary loyalty initiatives (e.g., VIP Pass)
- Prevent duplicate registrations and repeated welcome messages
- Enable QR code registration at events
- Automatically recognise returning beneficiaries and volunteers
- Encourage joining WhatsApp groups and following Facebook pages

5. Volunteer & Event Management

Current manual processes include:

- Volunteer registration
- Event communication
- Certificate generation
- Attendance tracking

The system should automate these workflows wherever possible.

6. GIFTIK Donation Drive

Potential areas for automation include:

- Automatic generation of event posters (PDF/JPG) by simply updating event date and time
- Emailing posters to predefined condominium management contacts

Current logistics workflow:

1. Coordinate collection drive with condominiums
2. Residents donate items
3. Arrange transport (LalaMove)
4. Deliver items to recreation centre
5. Volunteers sort items
6. Workers enter in batches (20–30 at a time)
7. Workers select 1–3 items
8. Event closes and volunteers clean up

Transportation itself does not require automation, as LalaMove already provides an effective solution.

7. Integrated Platform

Preferred solution:

- A **self-contained platform** with its own database.
- Eliminate dependence on Excel.
- Continue integrating with WhatsApp for communications.

8. Course Registration Challenge

Typical participation funnel:

- Large courses:
 - ~375 registrations
 - ~120 attend first session
 - ~90 complete
- Smaller courses:
 - ~100 registrations
 - ~30 attend
 - ~25–30 complete

Key observation:

- **Biggest drop-off occurs between online registration and first attendance.**
- Once participants attend the first session, completion rates are generally high.

This presents an opportunity for:

- Automated reminders
- Attendance confirmations
- Waitlists
- Better participant engagement before the first session

9. Stakeholder Management

Key external partners include:

- Ministry of Manpower (ACE Group)
- Recreation centre and dormitory management
- Condominium management
- Transportation providers (e.g., LalaMove)
- NGOs
- Educational institutions

Current communication is primarily via email.

Main challenge:

- Limited staffing leads to delays in responding to and coordinating with partners.
- Due to the diverse nature of these interactions, opportunities for automation are less clear and require further exploration.

Overall Vision

Develop a centralized digital platform (preferably mobile-first) that:

- Replaces Excel with a structured database
- Manages volunteers, beneficiaries, courses, events, and loyalty programmes
- Automates registrations, attendance, certificates, and communications
- Integrates with WhatsApp for messaging
- Supports QR-based onboarding and participant history
- Improves engagement, reduces manual administrative work, and scales across multiple beneficiary groups and programmes.

Templates

Volunteer

Standard Qns(First Timer Users):

Question 1: "Welcome! To get started, please reply with your full name." *(Text response)*

Question 2: "Thanks! Please share your contact email address." *(Text response)*

The Unified Identity Check

Question: "Welcome to Passion to Serve! To help us guide you, please let us know how you are joining us today:"

Poll Options:

-  I want to Volunteer
-  Migrant Worker
-  Low Income Household
-  Senior Citizen
-  Health / Disability Support
-  Social worker (Inmates/ Prisoners)

Volunteer Template :

Message 1 (Text):

"Awesome! Thank you for stepping up to serve. Here are the three ways you can get involved:

 **Items To Serve:** Help collect or distribute food, clothing, and daily essentials at our GIFTIK events, or assist with community beach cleanups.  **Knowledge To Serve:** Support our education programs by assisting in classes like digital literacy, coding, spoken English, or financial skills.  **Peace To Serve:** Help facilitate wellness activities, recreation, games, and yoga during our holiday events."

Message 2 (Poll):

"Which area would you like to volunteer for today?"

-  Items To Serve
-  Knowledge To Serve
-  Peace To Serve
-  Go Back

- **Path 1: If they select " Items To Serve"**

This path focuses on physical logistics, collections, and the e-platform.

Message:

"Great choice! Which 'Items To Serve' initiative would you like to support? Tap to select:"

Poll Options:

-  Food To Serve
-  Used Items Collection
-  Beach Cleanup
-  GIFTIK e-platform
-  Go Back

Role Preference: "What type of role do you prefer?"

Poll Options:

-  Logistics/Inventory
-  Registration
-  Usher/Crowd Control
-  Any role
-  Go Back

- **Path 2: If they select " Knowledge To Serve"**

Step 2A Message:

"Fantastic! Are you interested in providing general Academic Support, or specialized Learning & Enrichment?"

Poll Options:

-  Academic Support
-  Learning & Enrichment
-  Go Back

Step 2B Message (If they select Learning & Enrichment):

"Awesome! Which specific enrichment track would you like to teach or facilitate?"

Poll Options:

-  Financial / Digital Literacy
-  AI & Coding
-  Environmental (DIY Waste)
-  Arts & Crafts / Health
-  Go Back

Bridge Message: "Awesome! We have an upcoming *[Insert Track Variable]* session on *[Insert Date Variable]*. Let's get you registered!"

Role Preference: "What type of role do you prefer?"

Poll Options:

-  Lead Teacher
-  Assistant Facilitator
-  Tech Support
-  Any role
-  Go Back

- **Path 3: If they select " Peace To Serve"**

This path is straightforward and focuses on wellness and recreational events.

Message:

"Wonderful! Which 'Peace To Serve' activity would you like to help facilitate?"

Poll Options:

- 😊 Happiness Sessions
- 🎮 Recreation
- 🧘 Yoga/Mindfulness
- ⬅️ BACK Go Back

Shift Selection:

Message: "Awesome! We have an upcoming [Insert Activity Variable] event on [Insert Date]. Which shift are you available for? Tap to select:"

Poll Options:

- 🟢 Morning (9am-1pm)
- 🟠 Afternoon (1pm-5pm)
- 🔵 Full Day
- ⬅️ BACK Go Back

Dietary Needs (if meals are provided):

- *Question:* "Any dietary restrictions for lunch?"
- *Poll Options:* None | Halal | Vegetarian | Vegan | Other (Will message organizer)

System Closing Message: "Thank you for volunteering with us! If you are unable to make it closer to the date, reply 'CANCEL' to cancel your volunteering session. If you would like to volunteer for another session, reply 'VOLUNTEER'."

Beneficiaries

Beneficiary Template

"Hello! We are so glad you are joining us. What would you like to register for today?"

- **Poll Options:**
 -  Collect Daily Items / Food
 -  Join a Class / Enrichment
 -  Join a Wellness / Recreation Event
 -  Go Back

Path 1: Collect Daily Items / Food

- **Items Option (Multi-Select):** "What kind of items do you need the most? (You can select up to 3)"
 - **Poll Options:**  Dry Rations/Groceries |  Clothing |  Toiletries/Hygiene |  School Supplies |  Toys/Children's Items
- **Collection Timing:** "What time will you be coming down to collect your items?"
 - **Poll Options:**  10:00 AM - 11:30 AM |  11:30 AM - 1:00 PM |  1:00 PM - 2:30 PM
- **Accessibility Needs:** "Do you require any physical assistance during collection?"
 - **Poll Options:**  Yes, please |  No, I'm good

Path 2: Join a Class / Enrichment

- **Category Selection:** "Which educational track would you like to register for?"
 - **Poll Options:**  Academic Support |  Learning & Enrichment |  Go Back
- **Specific Class (If Learning & Enrichment is chosen):** "Which specific enrichment class would you like to join?"
 - **Poll Options:**  Financial/Digital Literacy |  AI & Coding |  Environmental (DIY Waste) |  Arts & Crafts/Health |  Go Back

Path 3: Join a Wellness / Recreation Event

- **Activity Selection:** "Which activity would you like to join?"
 - **Poll Options:**  Happiness Sessions |  Recreation |  Yoga/Mindfulness |  Go Back
- **Accessibility Needs:** "Do you require any physical accommodations for this activity?"
 - **Poll Options:**  Yes, please |  No, I'm good

Final Closing (All Paths)

- **System Closing Message:** "Thank you for registering! We look forward to seeing you. If you need to cancel your visit, simply reply 'CANCEL'." If you would like to join another event, reply 'JOIN'.

Events Partner

Step 1: Organization Identification

"Welcome, Partner! Thank you for collaborating with Passion to Serve. Are you a existing or new partner interested in a collaboration?"

- **Poll Options :**

-  Existing Partner
-  New Partner (Explore Collab)

Which organization are you representing today?"

- **Poll Options (Up to 12):**

-  MOM (ACE Division)
-  MINDS
-  Willing Hearts
-  Temasek Polytechnic
-  RSVP Singapore
-  UWCSEA
-  365 Cancer Prevention
-  The BARN (Cornerstone)
-  Doctors For You
-  **New Partner (Explore Collab)**
-  Go Back

Path 4: If they select " New Partner (Explore Collab)"

(Since they are not in the system yet, you need to collect unstructured data (text) alongside some structured data to qualify the lead.)

Step 4A: Organization Name

- **Message:** "We are thrilled to explore a partnership with you! To start, please type the full name of your organization." (*Text response*)

Step 4B: Primary Contact

- **Message:** "Thank you! Please type your name and your role at the organization (e.g., Jane Doe, Community Outreach Lead)." (*Text response*)

Step 4C: Area of Interest

- **Message:** "How are you hoping to collaborate with Passion to Serve?"
- **Poll Options:**
 -  Bringing Beneficiaries
 -  Providing Volunteers
 -  Sponsorship / Logistics
 -  Joint Event / Other
 -  Go Back

Step 4D: New Partner Closing

- **Message:** "Thank you for your interest! We have securely logged your details. Our partnership team will reach out to you via email or WhatsApp within 1-2 business days to discuss how we can serve together. Have a great day!"

Step 2: Purpose of Registration "Thanks for confirming! How are you partnering with us for the upcoming session?"

- **Poll Options:**
 -  Registering Beneficiaries
 -  Coordinating Volunteers
 -  Providing Logistics/Items
 -  Go Back

Step 3A: If " Registering Beneficiaries"

- **Message:** "Wonderful. Which pillar are your beneficiaries participating in today?"
- **Poll Options:**  Items To Serve |  Knowledge To Serve |  Peace To Serve |  Go Back
- *(Routing Logic: Once they select a pillar, the bot can seamlessly push them into the specific paths we built in the Beneficiary Template to select the exact class, activity, or item needed.)*

Step 3B: If " Coordinating Volunteers" or " Providing Logistics/Items"

- **Message:** "Great! Which pillar is your organization supporting today?"
- **Poll Options:**  Items To Serve |  Knowledge To Serve |  Peace To Serve |  Go Back

- *(Routing Logic: Once selected, push them into the specific Volunteer Paths to capture the exact track and shift timings.)*

Step 4: Group Size (Crucial for Partners)

- **Question:** "To help us prepare, how many individuals (beneficiaries or volunteers) will you be bringing/registering? Please type a number (e.g., 5, 12, 20)." *(Text response)*

Final Closing (All Partner Paths)

- **System Closing Message:** "Thank you for registering! We have logged your details. Our core team will review this and reach out shortly in 1-2 business days to discuss further."

outline for slides

Rough flow

#	Slide	Judging criteria it scores	What must be on the slide
1	Title + one-line thesis	Relevance (first impression)	Team name, thesis line: "WhatsApp is already the system — we gave it a brain and a front desk."
2	The problem, in their words	Relevance	3–4 direct quotes from Passion To Serve's own Q&A doc (Excel breaking down, staff shortage, drop-off rate).
3	Why past attempts don't fix this	Relevance + Creativity	Pre-empt the "why not just build X" question: new app = adoption friction; CRM = still manual relay; broadcast tool = violates Meta policy.
4	Our approach — Keep / Connect / Control	Relevance + Effectiveness	The three-word framing. Should be memorable enough a judge repeats it back.
5	Architecture	Technical Design & Code Completeness	Diagram: Form → Sheet → Dashboard ↔ GreenAPI ↔ WhatsApp ↔ Webhook. One source of truth, not three synced systems.
6	Feasibility for this org specifically	Effectiveness & Feasibility	No new app, no new account types beyond Forms/Sheets/GreenAPI, editable by anyone who can use a spreadsheet.
7	Live demo / skit (plan more indepth!)	Technical Design + Effectiveness	Real registration → WhatsApp confirmation → CANCEL → auto-withdrawal loop actually firing.
8	Feature: two-way thread, not broadcast	Creativity & Innovation	Webhook catching an inbound reply and mutating state — hard to fake in a demo, and it's real.
9	Feature: organiser-controlled messaging	Creativity + Social Impact	Editable templates, human-in-the-loop send — framed as a trust decision, tied to the org's stated values.
10	Social impact, quantified	Social & Environmental Impact	Translate the 30–40% show-up rate into what this fixes: real-time waitlist backfill, better-resourced events.
11	What we didn't build, on purpose	Effectiveness + Creativity	Name excluded scope (new app, full CRM rebuild, unreviewed AI messages) and why — restraint reads as maturity.

12	Roadmap	Social Impact + Feasibility	Volunteer loyalty scoring, QR check-in, extending the pattern to condo/transport partner coordination.
13	Close	Relevance	Return to the one-line thesis. No new information.

2.1 Relevance

Passion To Serve's own operational Q&A identifies three specific, named failure points, not a vague "we need digital transformation":

- Excel cannot track relationship history across events — explicitly cited as blocking their planned Volunteer Loyalty Programme and the recent VIP Pass initiative for GIFTIK.
- WhatsApp's Meta broadcast restrictions block bulk outbound communication to registered participants at scale, forcing manual one-to-one messaging.
- Chronic short-staffing means "time delays in interacting with stakeholders" — a capacity problem, not a tooling-absence problem.

The product directly targets all three: a Sheet as durable relationship history, GreenAPI's transactional (not bulk-broadcast) message pattern to work within WhatsApp's policy, and templated-but-editable bulk sends to reduce the manual burden without removing the organiser from the loop. Nothing in the product invents a need the org didn't already name.

2.2 Effectiveness & Feasibility

Effectiveness

The core mechanism — Form intake → Sheet as system of record → GreenAPI send/receive → webhook state sync — closes the specific loop the org described as broken: registration was previously one-directional (a Google Form that never got reconciled against who actually showed up). Making cancellation as easy as replying "CANCEL" to a message already in the person's inbox removes the single largest friction point in real intent-signaling — no login, no separate app, no new muscle memory.

Feasibility for this specific org

This is the more important test, and it's a stronger case than raw technical feasibility:

- No new account types beyond a Google Form (already familiar — currently used for registration), a Google Sheet (already their de facto database), and a GreenAPI WhatsApp instance.
- No dedicated engineering hire required to operate it post-hackathon — the org can edit questions and message templates without touching code.

Matches their own stated preference:

"we would prefer a self-contained solution however... we can do away with Excel completely if the solution itself provides robust database handling features"

The Sheet-as-database approach is a deliberate concession to their operating reality, not a technical shortcut being hidden.

Feasibility risk and mitigation

Risk	Mitigation
Google Sheets API rate limits at scale (>100 concurrent writes)	Event sizes described in their own numbers (up to ~375 registrations) are well within Sheets API quotas; no realistic near-term event volume threatens this.
GreenAPI account suspension for policy violation	Messages are transactional (tied to a specific registration action with opt-in), not cold broadcast — the pattern WhatsApp's business policy is built to allow.
Non-technical staff turnover losing institutional knowledge	Everything lives in a Sheet + Form, both already familiar to any new volunteer coordinator, unlike a bespoke app/database only the original builders understand.

2.3 Technical Design & Code Completeness

The system deliberately has one source of truth (the Sheet) rather than three systems that need reconciliation. This is a real architectural decision, not a default: the alternative — a dashboard with its own database, synced separately to Sheets for the org's comfort — would introduce a sync-failure class of bugs that doesn't exist here. Every component (Form, dashboard, GreenAPI send, inbound webhook) reads and writes the same row-level data.

The inbound webhook (GreenAPI → Lambda → Sheet row match → status mutation) is the piece of technical design worth defending explicitly to judges: it's what separates this from the far more common "send templated WhatsApp messages" hackathon submission, which is one-directional and doesn't close the loop on actual attendance intent.

2.4 Creativity & Innovation

The innovation here isn't inventing a new channel — it's recognizing that the org's stated constraints (no new app, keep WhatsApp, staff shortage) point toward integration rather than replacement as the creative answer. Two specific mechanisms carry the innovation claim:

- **Two-way state sync via a plain-language keyword ("CANCEL")** — Turns a one-way broadcast tool into a live registration state machine, without requiring the beneficiary to learn any new interaction pattern.
- **Editable-but-templated bulk messaging** — Resolves the tension between "we need to message hundreds of people" and "messages should still sound like a human wrote them," a tension most bulk-messaging tools resolve by picking one side (fully manual or fully automated).

2.5 Social & Environmental Impact

- Directly serves the org's own stated beneficiary base — migrant workers (currently ~1.64 million foreign workforce in Singapore, per their own deck), plus the low-income households, inmates, and broader community groups they've said they want to expand toward.
- Reduces the show-up gap. Their own numbers show only 30–40% of online registrations convert to actual attendance. Earlier, structured cancellation signals mean the org can backfill from a waitlist instead of over- or under-preparing for an event — GIFTIK events depend on volunteers sorting and laying out a fixed quantity of donated items in a ~1.5 hour window before doors open, so accurate headcounts materially affect how well that window is used.
- Strengthens, rather than routes around, human relationships. Every message is organiser-approved before sending — the product deliberately doesn't remove the human from beneficiary-facing communication, which matters for an org whose

stated identity includes "Ethics, Transparency, Accountability to our donors/volunteers/beneficiaries."

- Builds toward capability they already wanted but couldn't get from Excel — the Volunteer Loyalty Programme and VIP Pass tracking they described as blocked by their current tools become straightforward once relationship history lives in one clean, queryable Sheet instead of being re-entered per event.